

Value Extinction Threshold Module (VXTM)

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Canonical Definition

Value Extinction Threshold Module (VXTM) is the second Core Module of the Value Extinction Standard (VXSS) within VFM® — Value Flow Mechanisms Architecture.

VXTM governs the definition, application, testing, and documentation of the threshold separating value that remains legitimately recoverable from value whose recoverability has become sufficiently exhausted, unavailable, or terminated to justify formal Value Extinction determination.

The Module receives qualified extinction conditions from VXCM — Value Extinction Conditions Module and evaluates whether those conditions collectively reach the governed boundary beyond which restoration of the same relevant value can no longer be treated as a legitimate continuation pathway within the defined value lifecycle.

Its operational purpose is:

to establish whether the governed boundary between recoverable value loss and potential terminal non- recoverability has been reached before any formal Value Extinction determination is made.

Architectural Position

VXTM occupies the critical boundary between:

Extinction Conditions

and
Extinction Determination.

Its architectural position is:

VXCM — Value Extinction Conditions Module → VXTM — Value Extinction Threshold Module
→ VXDM — Value Extinction Determination Module

VXCM establishes which conditions are materially relevant.

VXTM establishes whether those conditions reach the required terminal boundary.

VXDM subsequently determines whether the relevant value is legitimately extinct.

Therefore:

Condition ≠ Threshold.

Threshold ≠ Determination.

Internal Flow

Governed Extinction Conditions → Relevant Value Boundary Confirmation → Recoverability Criteria Definition →
Restoration Boundary Definition → Threshold Criteria Definition → Threshold Evidence Association → Alternative
Recovery Pathway Test → Residual Recoverability Test → Threshold Satisfaction Assessment → Threshold Status →
Governed Extinction Threshold Result → VXDM — Value Extinction Determination Module

Operational Space

VXTM governs the space between:

material conditions threatening recoverability

and

formal determination that the relevant value has become extinct.

Its operational space includes:

recoverability criteria; restoration boundaries; extinction-threshold criteria; threshold evidence; threshold dependencies;
threshold satisfaction; residual recoverability; alternative recovery pathways; partial threshold satisfaction; dimensional
thresholds; contextual thresholds; temporal thresholds; compound thresholds; uncertainty; threshold reassessment;
threshold status.

VXTM does not itself declare Value Extinction.

It establishes whether the required boundary for such a determination has been reached.

The Extinction Threshold

The Value Extinction Threshold is the governed boundary separating:

value that remains recoverable within its defined lifecycle

from

value for which legitimate recoverability has been sufficiently exhausted to permit formal extinction determination.

The threshold must be connected to:

the specific governed value; the relevant Value Object; the relevant value dimension; the relevant Value Context; the
defined lifecycle; the restoration boundary; the available recovery pathways; the applicable recoverability criteria.

An extinction threshold cannot be meaningfully governed in abstraction from the value to which it applies.

Threshold ≠ Zero

The extinction threshold is not automatically reached when value reaches zero.

A value may enter:

ZVA™ — Zero Value Asset™

while retaining significant restoration potential.

Therefore:

Zero Value ≠ Extinction Threshold.

The zero-value condition describes the present Value State.

The extinction threshold addresses whether the pathway for recovery of that same value has reached its terminal boundary.

Threshold ≠ Nullification

DeMonet™ may move positive value into a zero-value condition.

That transition does not establish that the value is unrecoverable.

The architecture therefore preserves:

Positive Value → DeMonet™ → ZVA™ → Recoverability Remains → ReMonet™ remains possible

and separately:

ZVA™

→ Extinction Conditions

→ Extinction Threshold Reached

→ Extinction Determination

Therefore:

Value Nullification ≠ Value Extinction Threshold.

Recoverability Criteria

Before an extinction threshold can be applied, the meaning of recoverability must be sufficiently defined.

Recoverability criteria may include, where relevant:

technical recoverability; functional recoverability; informational recoverability; operational recoverability; relational recoverability; contextual recoverability; temporal recoverability; legal or rights-dependent recoverability; economic recoverability; structural recoverability.

Not every criterion must apply to every value.

The applicable criteria must derive from the nature of the governed value and its defined restoration boundary.

Recoverability Is Value-Specific

A Value Object may contain multiple forms or dimensions of value.

The extinction threshold must therefore apply to the relevant value rather than automatically to the entire object.

For example, a production machine may permanently lose:

productive value

while retaining:

material value

component value

historical value

or

informational value.

Therefore:

Threshold Reached for One Value Dimension $\neq$ Threshold Reached for Every Value Dimension.

Restoration Boundary

VXTM requires a defined restoration boundary.

The restoration boundary identifies the legitimate space within which recovery of the same value may still be treated as Value Restoration rather than creation of a different value.

This distinction is essential.

Without it, almost any future value associated with an object could incorrectly be interpreted as restoration of the original value.

The restoration boundary may include:

identity of the prior value; value dimension; value relationship; relevant context; functional continuity; temporal continuity; permissible transformation; restoration mechanism; required dependencies; acceptable discontinuities.

Same Value Test

VXTM must distinguish recovery of the same relevant value from creation of another value.

For example:

A destroyed industrial machine is replaced by a new machine.

Production capability returns.

But the new machine's productive value may represent:

New Value Emergence

rather than:

restoration of the extinguished productive value of the original machine.

Therefore:

Replacement Value $\neq$ Restored Value.

The extinction threshold concerns recoverability of:

the same governed value

not whether some substitute value can eventually perform a similar function.

Alternative Recovery Pathway Test

Before the extinction threshold is treated as satisfied, materially credible alternative restoration pathways should be considered.

Potential pathways may include:

repair; reconstruction; remanufacturing; technological conversion; data recovery; reconstruction from secondary sources; reactivation; rights recovery; contextual transfer; dependency replacement; controlled substitution where continuity of the same value remains legitimate.

The existence of one failed pathway does not establish threshold satisfaction.

Failed Recovery Pathway $\neq$ Exhausted Recoverability.

Residual Recoverability

Recoverability does not need to be complete to remain material.

A value may retain:

partial recoverability; conditional recoverability; dimensional recoverability; contextual recoverability; delayed recoverability; uncertain but credible recoverability.

Where such recoverability remains legitimately material:

the extinction threshold may not yet be satisfied.

Restoration Potential $\neq$ Restoration Certainty

A value does not need to be easily recoverable to remain below the extinction threshold.

The architecture distinguishes:

Restoration Potential

from

Restoration Certainty.

A difficult, expensive, technically complex, or uncertain restoration pathway may still represent legitimate recoverability.

Therefore:

Low Probability of Restoration $\neq$ Automatic Extinction Threshold.

Economic Threshold Principle

Current restoration cost may exceed current economic value.

That fact may influence organizational decisions.

But it does not universally establish Value Extinction.

For example:

Restoration Cost > Current Economic Return

may justify:

Do Not Restore

without establishing:

Cannot Be Restored as the Same Value.

Therefore:

Economic Non-Viability $\neq$ Universal Non-Recoverability.

Where economic recoverability is itself the governed value dimension, however, economic criteria may legitimately form part of the extinction threshold.

Technical Possibility $\neq$ Unlimited Recoverability

The opposite error must also be avoided.

The mere theoretical possibility that some future technology might recover value does not require VFM® to preserve the lifecycle indefinitely.

VXTM operates within a governed restoration boundary.

Recoverability should therefore be evaluated against:

established capabilities; reasonably identifiable pathways; materially credible dependencies; applicable governance criteria; known constraints; defined lifecycle assumptions.

Therefore:

Theoretical Future Possibility $\neq$ Established Restoration Pathway.

Threshold Criteria

A Value Extinction Threshold may require one or more criteria to be satisfied.

Examples may include:

essential value basis irreversibly lost; required value relationship permanently terminated; critical information irrecoverably unavailable; required capability no longer reconstructable; all legitimate restoration pathways exhausted; necessary dependencies permanently unavailable; restoration would create fundamentally new value rather than recover the prior value; contextual conditions permanently eliminate the relevant value relationship; temporal boundary permanently passed; required continuity irreversibly broken.

The applicable threshold must remain specific to the governed value.

Single-Criterion Threshold

Certain value relationships may depend upon one indispensable condition.

If that condition is irreversibly lost, a single criterion may be sufficient to satisfy the extinction threshold.

For example:

A unique cryptographic key may be indispensable to a particular access value.

If the key is irretrievably lost and no legitimate recovery or reconstruction pathway exists, the loss of that single dependency may be sufficient to reach the threshold for that specific access value.

Compound Threshold

Many extinction cases require multiple criteria.

A Compound Extinction Threshold exists where several conditions collectively establish terminal non-recoverability.

For example:

Critical Component Loss

Source Documentation Loss

Control-System Incompatibility

No Reconstruction Capability

No Alternative Restoration Pathway

may collectively satisfy the threshold even where no single condition would independently be terminal.

Necessary and Sufficient Criteria

VXTM may distinguish:

Necessary Criteria

from

Sufficient Criteria.

A necessary criterion must be present before the threshold can be reached.

A sufficient criterion, where legitimately defined, may independently establish threshold satisfaction.

Compound thresholds may require:

Criterion A + Criterion B + Criterion C

before terminal-boundary satisfaction occurs.

This structure prevents arbitrary threshold interpretation.

Threshold Materiality

Threshold criteria must be materially connected to recoverability.

A condition should not contribute to threshold satisfaction merely because it is negative, inconvenient, costly, or undesirable.

The relevant question remains:

Does this criterion materially affect whether the same governed value can legitimately be recovered?

Threshold Satisfaction States

VXTM may recognize:

1. Threshold Not Reached

Material recoverability remains.

2. Threshold Approaching

Extinction conditions are materially reducing recoverability, but the terminal boundary has not been reached.

3. Threshold Conditionally Reached

The threshold is satisfied only under defined contextual, temporal, or dependency conditions.

4. Threshold Partially Reached

The threshold is satisfied for one value dimension or portion of the governed value but not others.

5. Threshold Reached

Applicable extinction-threshold criteria have been satisfied sufficiently to permit formal Value Extinction determination.

6. Threshold Indeterminate

Available information is insufficient to establish whether the threshold has been reached.

These states preserve uncertainty and prevent binary conclusions where evidence does not support them.

Threshold Reached ≠ Value Extinct

This is the central governance boundary of VXTM.

Reaching the threshold means:

the case is eligible for formal Value Extinction determination.

It does not mean:

Value Extinction has already been formally determined.

Therefore:

Threshold Reached ≠ Extinction Determined.

The formal determination belongs to:

VXDM — Value Extinction Determination Module.

Partial Threshold Satisfaction

A Value Object may cross the extinction threshold for only part of its value.

For example:

An industrial system may permanently lose its original production capability while retaining valuable materials and technical documentation.

VXTM may therefore establish:

Productive Value → Threshold Reached

while:

Material Value → Threshold Not Reached

and:

Informational Value → Recoverability Retained.

Contextual Threshold

The threshold may be reached within one context without being reached universally.

A technology may become permanently unusable under one regulatory, technical, or operational environment while remaining usable elsewhere.

Therefore:

Contextual Threshold Reached ≠ Universal Value Extinction.

The context must remain explicit.

Temporal Threshold

Certain value depends upon a time-bound opportunity.

Where the relevant temporal boundary passes irreversibly, the associated value may become unrecoverable.

However, VXTM must distinguish:

loss of a specific historical opportunity

from

loss of every future value associated with the underlying object.

The expired opportunity may cross its extinction threshold while other value persists.

Threshold Uncertainty

Where material uncertainty remains concerning:

alternative restoration pathways; condition reversibility; missing evidence; technological recoverability; dependency reconstruction; continuity of the same value;

VXTM should preserve that uncertainty.

A threshold should not be declared reached merely because recovery has not yet been demonstrated.

Absence of Demonstrated Recovery ≠ Demonstration of Non-Recoverability.

Threshold Reassessment

Material new information may justify reassessment before formal extinction determination.

Examples include:

discovery of backup data; identification of an alternative repair pathway; recovery of missing documentation; discovery of a previously unknown dependency; new evidence that a condition is reversible; discovery that the relevant value dimension was incorrectly defined.

VXTM therefore permits threshold reassessment while the lifecycle remains open.

Boundary with VXCM

VXCM asks:

What conditions materially threaten recoverability?

VXTM asks:

Have those conditions reached the governed terminal boundary?

Therefore:

Extinction Conditions $\neq$ Extinction Threshold.

Boundary with VRSS

VRSS governs the pathway through which previously existing value may be restored.

VXTM governs whether legitimate restoration potential has become sufficiently exhausted to permit extinction determination.

The architecture therefore creates a decisive branch:

Recoverability Retained $\rightarrow$ VRSS — Value Restoration Standard

Recoverability Terminally Exhausted $\rightarrow$ VXSS — Value Extinction Pathway

Boundary with VNS

VNS governs the transition to zero value.

VXTM governs whether recoverability of that zero value has reached its terminal boundary.

Therefore:

ZVA™ may remain below the extinction threshold.

Boundary with VXDM

VXTM determines threshold status.

VXDM determines extinction status.

Therefore:

Threshold Assessment $\neq$ Extinction Determination.

Minimum Implementation Framework

A minimum VXTM implementation should establish:

1. Governed Value Boundary

Confirm the exact value, dimension, context, and lifecycle under assessment.

2. Extinction Condition Input

Receive qualified Extinction Conditions from VXCM.

3. Recoverability Criteria

Define what constitutes legitimate recovery of the same value.

4. Restoration Boundary

Define the boundary separating restoration from replacement, substitution, or New Value Emergence.

5. Threshold Criteria

Define necessary, sufficient, compound, contextual, dimensional, or temporal threshold criteria.

6. Alternative Pathway Test

Assess materially credible restoration pathways.

7. Residual Recoverability Test

Determine whether partial, conditional, contextual, or other material recoverability remains.

8. Threshold Satisfaction Assessment

Evaluate the governed conditions against the applicable threshold.

9. Threshold Status

Assign the appropriate threshold state.

10. Determination Transfer

Transfer the governed threshold result to VXDM for formal Value Extinction determination.

Governance Outputs

VXTM may produce:

Value Recoverability Criteria Record Value Restoration Boundary Value Extinction Threshold Definition Threshold Criteria Record Alternative Recovery Pathway Assessment Residual Recoverability Assessment Compound Threshold Assessment Dimensional Threshold Record Contextual Threshold Record Temporal Threshold Record Threshold Uncertainty Record Threshold Satisfaction Assessment Value Extinction Threshold Status Governed Extinction Threshold Result

Enterprise Application

VXTM provides organizations with a governed boundary for deciding when impaired, dormant, inaccessible, zero-value, or failed-restoration assets should move from recovery governance into formal extinction consideration.

This is especially relevant to:

industrial asset portfolios; legacy infrastructure; distressed investments; intellectual property portfolios; abandoned technologies; obsolete platforms; digital assets; historical data; discontinued products; stranded assets; long-term research programs; organizational capabilities.

Without such a threshold, organizations may make two opposite errors:

Premature Extinction — abandoning recoverable value.

or

Artificial Recoverability — indefinitely treating terminally lost value as recoverable.

VXTM creates the governance boundary between those two states.

Use Case 1 — Specialized Industrial Asset

A specialized industrial machine has reached ZVA™ after severe degradation.

The manufacturer no longer exists, essential components are unavailable, original control software has been lost, and two restoration attempts have failed.

VXCM has already established material Extinction Conditions.

VXTM now tests:

ZVA™ → Recoverability Criteria → Restoration Boundary → Component Reconstruction Possibility → Software Reconstruction Possibility → Alternative Modernization Pathway → Residual Productive Value Recoverability → Threshold Assessment

If a legitimate modernization pathway can restore the same productive value relationship, the threshold remains unreached.

If every legitimate pathway has been exhausted and any replacement would constitute New Value rather than restoration of the machine's prior productive value:

Productive Value Extinction Threshold → Reached

The machine may still retain material or historical value.

Use Case 2 — Irretrievably Lost Digital Access

Value

A digital asset depends upon a unique cryptographic credential.

The credential has been permanently lost.

No backup exists.

No recovery key exists.

No reconstruction pathway exists.

The underlying system remains operational, but the specific access relationship represented by the credential cannot be recreated as the same governed access value.

VXTM evaluates:

Credential Loss → Recoverability Criteria → Alternative Recovery Pathways → Dependency Reconstruction Test → Same Value Test → Residual Recoverability Test → Threshold Satisfaction

If a replacement credential would establish a fundamentally new access relationship rather than restore the prior one:

Original Access Value Extinction Threshold → Reached

This permits the case to proceed to formal extinction determination without incorrectly declaring every other value associated with the digital asset extinct.

Foundational Principles

Zero Value ≠ Value Extinction Threshold.

Value Nullification ≠ Value Extinction Threshold.

Extinction Condition ≠ Extinction Threshold.

Extinction Threshold ≠ Extinction Determination.

Threshold Reached ≠ Value Extinct.

Failed ReMonet™ ≠ Extinction Threshold Reached.

Failed Recovery Pathway ≠ Exhausted Recoverability.

Low Restoration Probability ≠ Automatic Extinction.

Economic Non-Viability ≠ Universal Non-Recoverability.

Theoretical Future Possibility ≠ Established Restoration Pathway.

Replacement Value ≠ Restored Value.

Residual Recoverability must be tested before terminal threshold satisfaction.

Thresholds apply to specific value dimensions, contexts, relationships, and lifecycles.

A threshold may be reached for one value dimension while other value persists.

Absence of demonstrated recovery is not proof of non-recoverability.

The extinction threshold exists to determine when formal extinction determination becomes legitimate—not to make that determination itself.

Canonical Closing Statement

Value Extinction Threshold Module establishes the governed boundary between recoverable value loss and potential terminal non-recoverability within VFM®. Through explicit recoverability criteria, restoration boundaries, threshold criteria, alternative recovery pathway testing, residual recoverability assessment, dimensional and contextual boundaries, uncertainty preservation, and formal threshold status, VXTM prevents zero value, failed restoration, obsolescence, economic non-viability, or severe impairment from being automatically interpreted as Value Extinction. The Module determines when the governed basis for formal extinction determination has been reached while preserving the architectural distinction between Value Nullification, Value Restoration, and Value Extinction. Value crosses the extinction threshold not merely because recovery is difficult or absent today, but because the legitimate pathway for recovery of that same governed value has reached its defined terminal boundary.

Modul3

Governed Links

[→ Value Extinction Standard \(VXSS\)](#)

[→ VFM® Architecture Map](#)

[→ Previous: Value Extinction Conditions Module \(VXCM\)](#)

[→ Next: Value Extinction Determination Module \(VXDM\)](#)

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